



Term Evaluation (Odd) Semester Regular/Back Examination September 2026

Roll no. 2592554

Name of the Program: BCA AI&DS
Semester: 3rd
Name of the Course: Introduction to Machine Learning
Course Code: TBD303
Time: 1.5 hours

Maximum Marks: 25

Note:

- (i) This QP has 3 sections. Attempt questions from each Section.
- (ii) Answer the questions as per the instructions in each section.

Section A

This section has MCQ's. Attempt any 5. Each question carries 1 mark.

Question 1.

a. Which of the following best describes Machine Learning? (1 Mark) CO1

- A. A technique used only for database management
- B. A method that enables computers to learn patterns from data
- C. A programming language used for numerical computation
- D. A method used only for designing websites

b. Which type of Machine Learning uses labelled data for training? (1 Mark) CO1

- A. Unsupervised Learning
- B. Reinforcement Learning
- C. Supervised Learning
- D. Semi-supervised Learning

c. Which Python library is primarily used for numerical and array-based computations in Machine Learning? (1 Mark) CO2

- A. NumPy
- B. Flask
- C. Tkinter
- D. BeautifulSoup

d. Which of the following is an example of a regression problem? (1 Mark) CO1

- A. Predicting whether an email is spam or not
- B. Grouping customers into different segments
- C. Predicting the price of a house
- D. Classifying an image as a cat or dog

e. In Linear Regression, the dependent variable is generally represented as a function of: (1 Mark) CO2

- A. Independent variable(s)
- B. Clusters only
- C. Class labels only
- D. Rewards only



Term Evaluation (Odd) Semester Regular/Back Examination September 2026

- f. Which algorithm is commonly used for classification based on the nearest training examples? (1 Mark) CO2
- A. k-Nearest Neighbors
 - B. K-Means
 - C. PCA
 - D. Apriori
- g. Which algorithm classifies an observation based on the classes of its nearest neighbours? (1 Mark) CO2
- A. Linear Regression
 - B. k-Nearest Neighbors
 - C. Polynomial Regression
 - D. SVM
- h. Which of the following is a supervised classification algorithm? (1 Mark) CO2
- A. K-Means
 - B. PCA
 - C. Decision Tree
 - D. Apriori

Section B

Attempt any 2 sub-question out of given 4 questions. Each question carries 5 marks.

Question 2

- a. Define Machine Learning and explain its scope. Discuss Supervised, Unsupervised, Semi-supervised, and Reinforcement Learning with suitable examples. (5 Marks) CO1
- b. Explain the major applications of Machine Learning in different domains such as healthcare, finance, education, recommendation systems, and cybersecurity. (5 Marks) CO1
- c. Explain the role of PyTorch, NumPy, Pandas, and Scikit-learn in developing a Machine Learning project. Also, briefly discuss TensorFlow/Keras. (5 Marks) CO1
- d. Differentiate between Supervised Learning and Unsupervised Learning on the basis of training data, objective, output, algorithms, and applications. (5 Marks) CO1

Section C

Attempt any 2 question out of given 4 questions. Each question carries 5 marks.

Question 3

- a. Explain Linear Regression with its mathematical equation. Describe the meaning of dependent variable, independent variable, slope, and intercept with a suitable example. (5 Marks) CO2
- b. What is Polynomial Regression? Explain how polynomial regression differs from linear regression and give a suitable application example. (5 Marks) CO2
- c. Explain Logistic Regression and describe the role of the sigmoid function in binary classification. (5 Marks) CO2
- d. Explain the working of k-Nearest Neighbors (k-NN). Discuss the effect of the value of k on classification performance with a suitable example. (5 Marks) CO2